

Week 1

Thinking Like a Data Scientist

A useful analysis begins with a question, not with an algorithm.

Chapters 1-2 • Lecture

THE HABIT FOR TODAY

**Ask what one row means
before calculating anything.**

Every count, mean, groupby and merge
in this course starts with that sentence.

Two hours: from a vague concern to a defensible question

0-14	Where data science actually gets used — ten cases
14-24	What data is, our dataset, and how you are assessed
24-46	From a concern to a question: claims, confounding, the lifecycle
46-70	One row, one meaning — and the error that runs without warning
70-80	Break
80-110	Where the numbers came from: samples, proxies, joins, ethics
110-120	Diagnostic, exit ticket, and Lab 1 handoff

By the end of today you can

- turn a vague concern into a question data can answer
- say what kind of claim a piece of evidence supports, and what it does not
- read a table and say what one row of it is — before counting anything
- name one limitation of any number you report

★ OPENING QUESTION

When have you seen a number used to make a decision? Did the number deserve your trust?

Artificial intelligence, machine learning, data science

You will hear these four terms used as if they were the same thing. They are not.

Artificial intelligence

The broad goal: build systems that do things we associate with intelligent behaviour.

Machine learning

One part of AI: learn patterns from data in order to predict or decide.

Deep learning

One kind of machine learning, built from neural networks.

Data science — this course

Answering questions with data, saying how certain the answer is, and supporting a decision.

It may use machine learning. It also covers collecting data, cleaning it, summarising it, drawing it, testing it, and explaining it to someone who will act on it.

Most of the useful work in this field involves no model at all.

★ CHECK YOUR UNDERSTANDING

Which is data science with no machine learning: predicting who will withdraw, or comparing satisfaction with a chart?

BEFORE ANYTHING ELSE

14 min

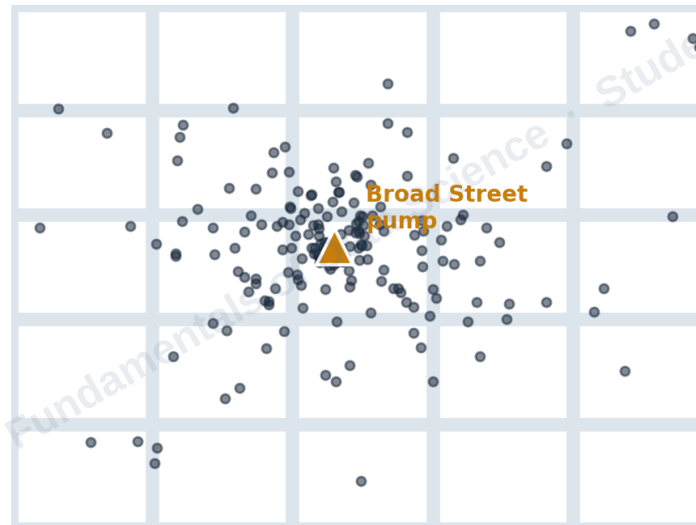
Where does this actually get used?

Ten real cases. Some changed policy, some cost lives, and one of them is about you.
Watch for what they have in common — we come back to it.

A map that ended an epidemic

Cholera was killing hundreds a week. The accepted explanation was bad air. John Snow did not argue about the theory — he plotted where the dead lived.

London, 1854: each dot is one death



The deaths clustered around one water pump. The handle was removed. The outbreak stopped.

NOTICE

He did not have statistics, software, or a model. He had a question, a unit of observation, and a map.

Snow's question had four parts — and so will yours

Strip away the century and the cholera, and the structure is the one you will use for the rest of this course.

Population	people living in Soho, autumn 1854	not “Londoners”, and not “the poor”
Variable	address of each death	not “how bad the air smelled” — he chose something countable
Period	the weeks of the outbreak	short enough that the cause had not changed
Decision	whether to remove the pump handle	a specific act, decided in advance

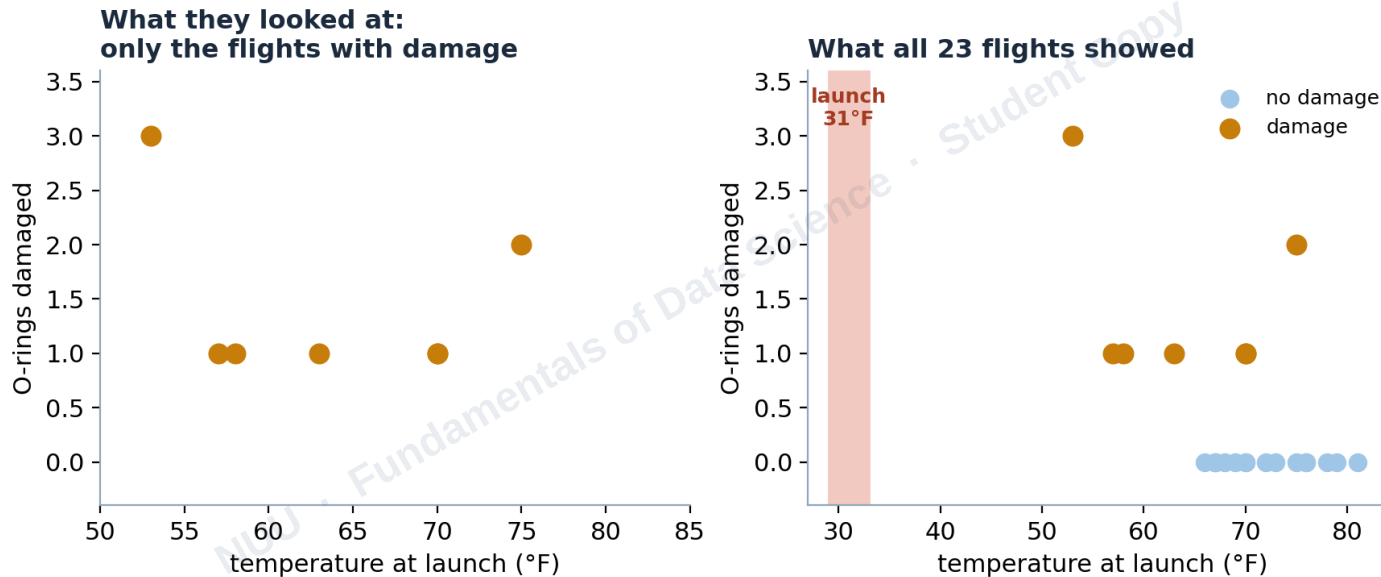
And one row of his data was one death. He knew that before he started counting.

KEEP THIS SHAPE

You will see these four words again in twenty minutes, applied to a request from your own Dean.

The chart that did not prevent a disaster

The night before launch, engineers argued the O-rings would fail in the cold. They showed the flights that had been damaged. Nobody plotted the flights that had not.



Left: the evidence presented. Right: the same evidence, complete. Launch temperature was 31°F — far colder than anything ever flown.

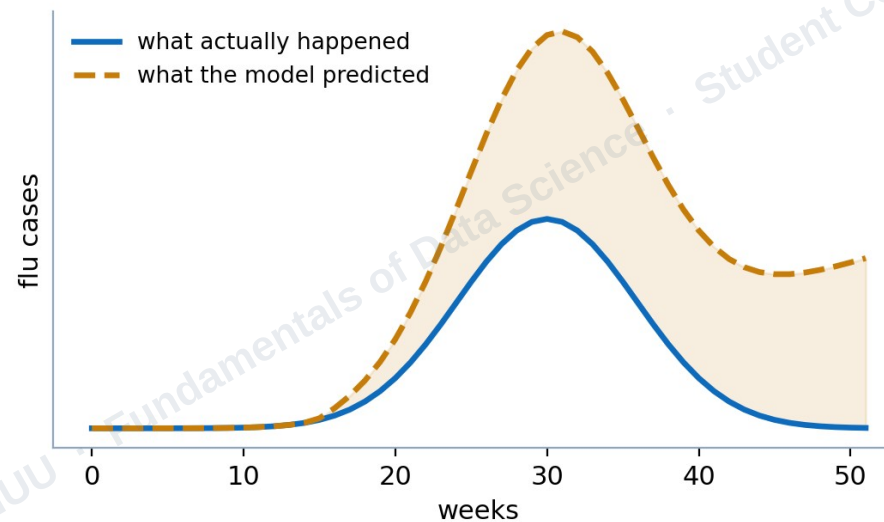
★ SIT WITH THIS

The data were correct. The analysis was not wrong. The picture was simply missing the half that mattered.

A model that worked, until it did not

Search queries were used to predict flu outbreaks faster than health agencies could report them. For two years it was celebrated. Then it began overshooting, badly and consistently.

A model can start well and quietly get worse



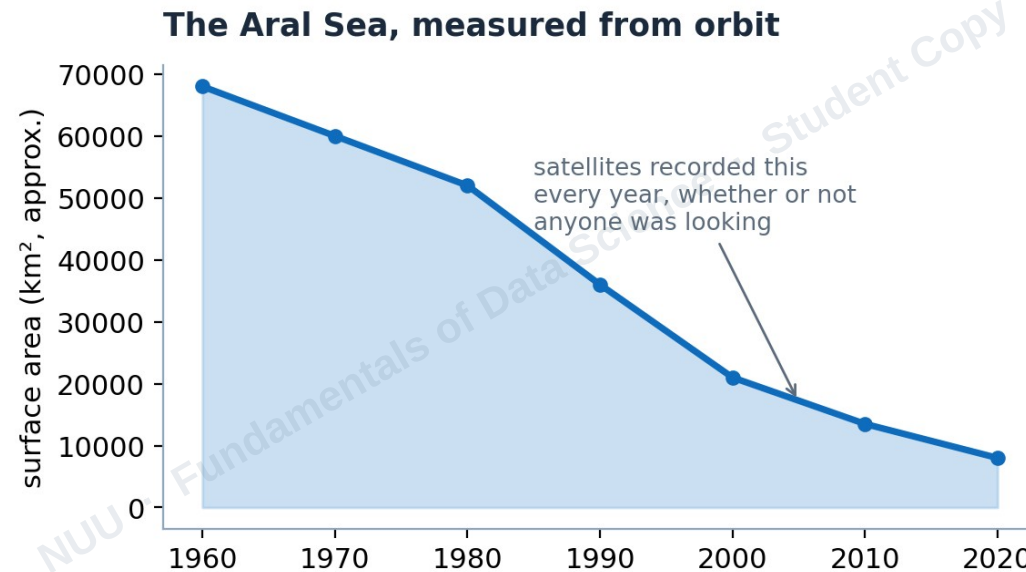
More data did not fix it. The world had changed underneath the model, and nobody was checking.

QUESTION TO HOLD

If a model is right for two years and then quietly stops being right, how would you find out?

Some data nobody chose to collect

No survey team measured this. Satellites photographed the same region on a schedule, and decades later those images became a dataset that could answer questions nobody had asked yet.



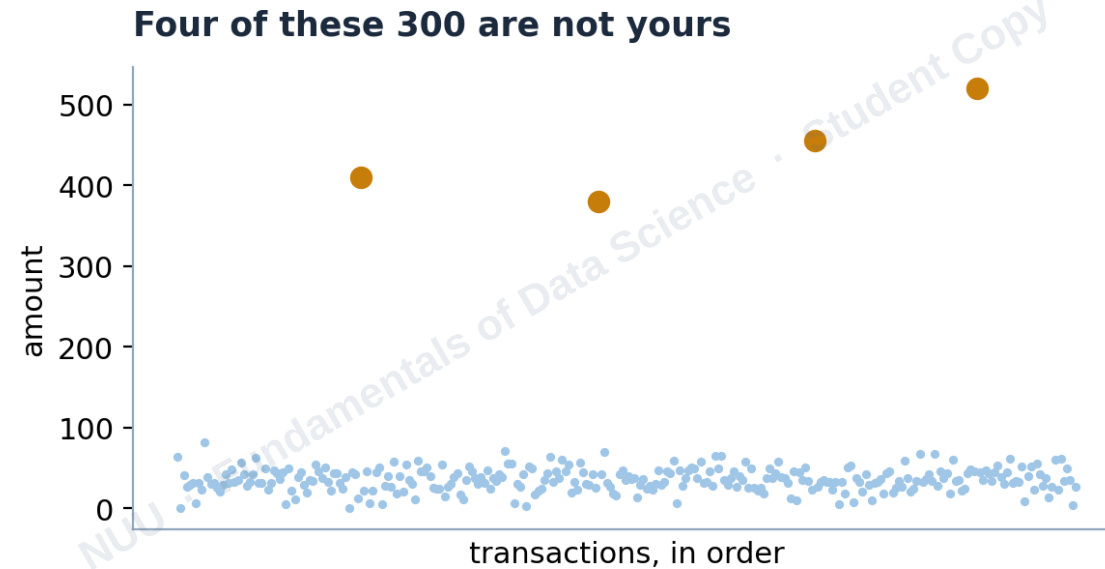
Approximate surface area. The data existed long before the question did.

NOTICE

The most useful datasets are often exhaust — records kept for some other purpose entirely.

The model that guards your bank card

Every card transaction is scored in milliseconds against what is normal for you. A handful get stopped.



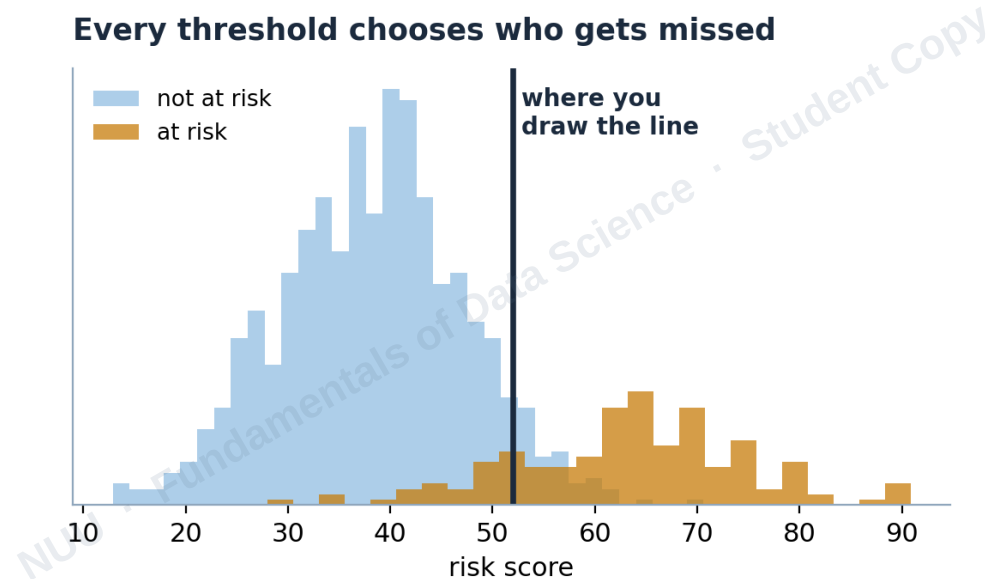
Stop too few and people lose money. Stop too many and a student is stranded at a till with a declined card.

THE REAL QUESTION

It is never “is this fraud?” It is “how sure do we need to be before we act?”

Someone has to draw the line

A screening score separates two groups that overlap. Wherever you put the line, some people at risk fall on the wrong side of it, and some people who are fine get called in.



Move the line left: more false alarms. Move it right: more missed cases. There is no position that avoids both.

★ THIS RETURNS IN WEEK 10

Choosing where the line goes is not a statistical decision. It is a decision about who bears the cost of being wrong.

Four systems, one shape

1 What you see next

A streaming or music service ranks thousands of items for you specifically. It is guessing from what people similar to you did next.

2 When you will arrive

A maps app predicts your journey time from millions of past journeys on the same road at the same hour — and updates while you drive.

3 What a search means

A translation or search tool learned patterns from enormous amounts of text. It has never understood a road or a word; it has seen how they are used.

4 Which crops need water

Satellite and sensor data estimate soil moisture across a whole region, so irrigation goes where it is needed rather than everywhere.

COMMON THREAD

Every one of them is a guess about something unobserved, made from records of what happened before.

A university asks: who is struggling, and how would we know?

Every institution holds records it never collected for research: registrations, submissions, logins, grades, survey responses.

The questions it wants answered

Are first-year students engaging less than last year?

Which programmes have the widest gap in outcomes?

Can we identify a student who is drifting, early enough to offer help?

Did the support we offered actually change anything?

The questions it should ask first

What does “engaging” mean, in a column?

Who is missing from these records entirely?

If we flag a student, what happens to them next?

Who is harmed if we are wrong — and can they argue back?

This is the dataset you will use every week of this course — 1,200 students, seven files, and every one of the problems above.

★ FOR THE NEXT TWO HOURS

We are the analyst the university just asked. Everything today is a step toward answering honestly.

Four moves, in every case

01

Somebody had a question about the world

not about a dataset, and not about a method

02

The data were a record of something else

a pump, a launch, a search box, a card reader, a registry

03

Someone decided what a row meant

one death, one flight, one week, one transaction, one student

04

The answer changed a decision

a handle, a launch, a policy, a payment, a student's term

Not one of them started with an algorithm. Three of the four moves happen before any code is written.

★ ROW 03 IS TODAY

Everything in the next two hours is about that third line, and it is the one people skip.

What is data? What is a table?

If you have used a spreadsheet, you already know most of this. The words below are the ones we will use all term.

student_id	programme	entry_score
100001	Computer Sci.	72
100002	Psychology	65
100003	Economics	58

← one row = one student

CSV file

A table saved as plain text, one line per row, values separated by commas.

Excel opens it. So does Python.

↑ one column = one attribute, measured the same way for everybody

A dataset is one or more of these tables, kept together, describing the same thing. Reading it into Python with pandas gives you a DataFrame — the same table, now something you can compute with.

ORIENTATION

Nothing today requires you to write code. Session 0 covers the Python you need — today, just read the table and ask what it is counting.

One university, 1,200 students, seven files

Every example in this course — today, and in Week 12 — uses these same students. Effort spent understanding them now keeps paying.

File	Rows	One row is
students.csv	1,200	one student
enrolments.csv	4,665	one student-course registration
activity.csv	16,165	one student-week on the platform
outcomes.csv	1,200	one student, early features plus withdrawal
feedback.csv	732	one free-text course comment
platform_daily.csv	364	one day of platform sessions
students_dirty.csv	1,205	students.csv with realistic damage

Where it came from

It is synthetic. No real student record is in it, and nobody's privacy is at stake.

But it was built to behave like a real registry export — including the flaws. Some of what you find in it will surprise you.

It lives in your Drive at
MyDrive/fds/data/

★ CHOOSE THE FILE

Which file would you open to answer: how many students are enrolled this year?

What this course expects from you

CLO1 — Explain

Data collection, preprocessing, exploratory analysis, and basic statistics.

CLO2 — Prepare

Numerical, categorical, time-series and text data: clean and transform appropriately.

CLO3-4 — Apply & communicate

Supervised and unsupervised methods, plus responsible visualisation, dashboards and reports.

CLASS NORM

A result is not complete until you explain what it means and name at least one limitation.

QUICK REFLECTION

Which assessment will test whether you can do data science — not merely repeat definitions?

Is your environment working? Hands up now

Open the Lab 0 setup cell and run it. You should see exactly this:

```
(1200, 7) (4665, 5) (16165, 6) (1200, 7)
```

students	1,200 rows · 7 columns
enrolments	4,665 rows · 5 columns
activity	16,165 rows · 6 columns
outcomes	1,200 rows · 7 columns

If you do not see this

Write your name on the sheet at the front before you leave today.

A student who is not set up in Week 1 is still not set up in Week 4 — and is behind on content as well.

The data folder belongs in Drive as:
MyDrive/fds/data/

★ DO IT NOW

Two minutes. Everyone runs the setup cell. I want names, not a show of hands I cannot follow up.

What your mark is made of

Component	Due	Weight
Midterm exam	Week 6	20%
Final exam	Exam period	25%
Weekly lab notebooks (best 9 of 12)	Weekly	15%
Data Quality Report	Week 4	10%
EDA and Visualization Brief	Week 7	10%
Modelling Assignment	Week 11	10%
Capstone project and dashboard	Week 12	10%

The exams are not definitions

Every question shows you something — a table, a result, five lines of code, a chart — and asks a judgement question about it.

No question asks you to define a term.

Memorising the glossary will not pass either paper. Doing the labs will.

QUICK REFLECTION

Which component tests whether you can do data science — not merely repeat it?

What each component is, and what you actually submit

Midterm & final exams	45%	Written, in person. Applied questions about tables, results, code and charts — no definitions.
Weekly lab notebooks	15%	One .ipynb per week, submitted in the LMS. Marked pass / revise / fail. Best 9 of 12 count.
Data Quality Report	10%	A notebook that finds the problems, plus a short written report saying what you did about each.
EDA & Visualization Brief	10%	A short document of charts. Each chart needs a title that states its point, and a named claim type.
Modelling Assignment	10%	A notebook with a model — and a written justification of where you set the threshold. That is 25 of its 100 marks.
Capstone + dashboard	10%	Your own public dataset, whole job end to end. The limitations section is 20 of its 100 marks.

★ FROM WEEK 4 ONWARD

Runtime → Restart and run all before you submit. If it only works in your own strange order, it does not count.

From a concern to a question

Requests arrive as sentences, not as questions. Repairing them is the first real skill, and the one most often skipped.

A concern is not yet an analytical question

The Dean says:

***“Students seem disengaged this year.
Can you look into it?”***

What is missing?

Who?

What does “disengaged” mean?

Compared with what?

What decision follows?

That sentence contains

no population
no variable
no time period
no decision

It is a real request from a real person, and it is the form most requests arrive in. Repairing it is the job.

★ THINK • PAIR • SHARE

Ninety seconds: rewrite this concern as a question that data could genuinely help answer.

Four parts make a question analysable

1

Population

Who is the conclusion about?

2026 entry cohort

2

Variables

What will you observe?

weeks 1–4 submissions

3

Period

When does it apply?

first four teaching weeks

4

Purpose

What decision follows?

target support by
programme

TRY IT

What population, variable, period and decision would you use to study “students are struggling with programming”?

Repair the concern in four passes

1. Who?

Restrict to the 2026 entry cohort — 431 students. Check the size first: a question about 40 people is a different question from one about 400.

2. What?

“Disengaged” is not a column. Choose an operational definition and state it: submissions in weeks 1–4.

3. Compared with what?

Against the 2025 cohort at the same point. Without a comparison there is no finding, only a number.

4. What decision?

If a difference concentrates in particular programmes, support is targeted rather than universal.

Repaired question: Among the 2026 cohort, is mean submission over weeks 1–4 lower than for 2025 at the same point, and does any difference concentrate by programme?

NAME THE RUNG

This question is comparative, not causal. What would you need to make a causal claim instead?

Not every question needs — or permits — the same evidence

What the data must support

Causal	Would mentoring reduce withdrawal?	<i>Randomisation or a strong design</i>
Predictive	Which early signals flag withdrawal?	<i>Held-out evaluation</i>
Comparative	Does it differ across programmes?	<i>Counts plus uncertainty</i>
Descriptive	What was last year's withdrawal rate?	<i>A clean count</i>

★ IMPORTANT DISTINCTION

Which rung is supported by a simple correlation between logins and grades?

Association is not an intervention result

“Students who log in more often get better grades, so the university should require weekly logins.”

Claim A — supported

Total logins and mean grade are positively associated: $r = 0.68$, $n = 1,200$.

Real, substantial, and descriptive. It sits on rung one.

Claim B — not supported

Requiring logins would raise grades.

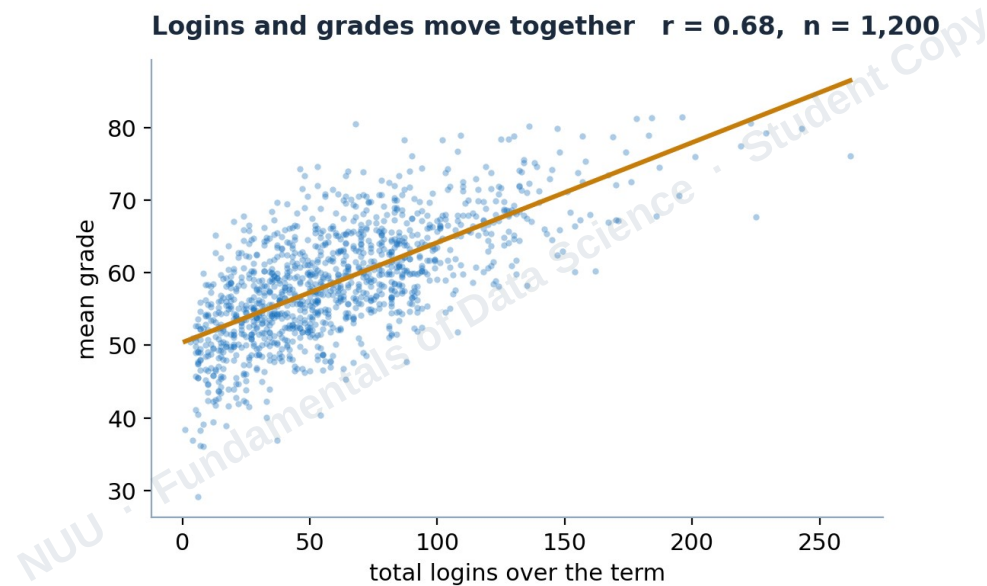
That is causal — rung four. Nothing in the correlation above supports it, and no amount of extra data will.

★ VOTE

Name one alternative explanation for the association besides “logging in causes learning.”

This is the association, drawn

1,200 students. Each dot is one of them: total logins across the term, against their mean grade.



The relationship is real and substantial. Nothing about the picture tells you which way the arrow points.

★ LOOK AGAIN

Can you see anything in this chart that rules out “good students log in more”?

A common cause fits the data just as well

Three explanations fit $r = 0.68$ equally well. The third is demonstrably present in this dataset.

Logins cause learning

Requiring logins would help.

Reverse direction

Students who understand the material find the platform rewarding and return to it.

A common cause

Prior preparation drives both. This is confounding — and we can check it.

entry_score, measured before anyone logged in at all, correlates with both:

0.402

entry_score with total logins

0.622

entry_score with mean grade

The honest sentence

“Entry score is associated with both, so this figure should not be read as an effect of logging in.”

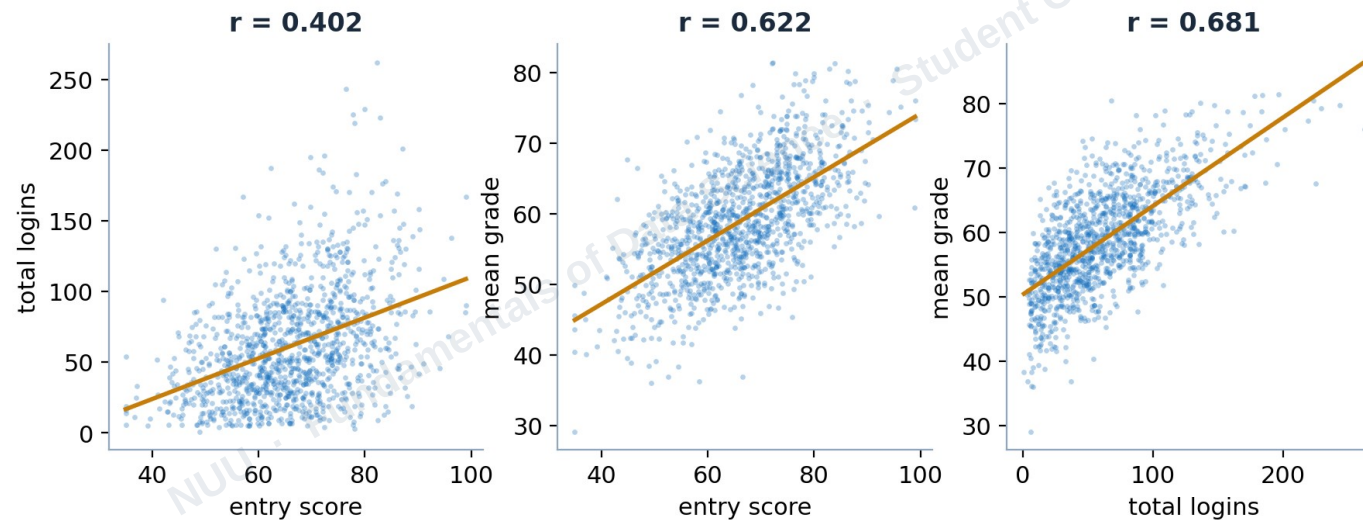
RESPONSIBLE USE

A mandatory-login policy would fall hardest on students with jobs or caring responsibilities. Who bears the cost if the claim is wrong?

One variable sits behind both

Entry score was recorded at admission — months before anyone touched the platform. It cannot have been caused by logging in.

Entry score was measured before anyone logged in — and it predicts both



Left and centre: entry score predicts logins, and predicts grades. Right: the association we started with — partly an echo of something that happened before term began.

THE HONEST SENTENCE

“Logins and grades are associated, but entry score predicts both, so this is not evidence that logging in causes learning.”

Good analysis loops; it does not move in a straight line



← Evaluation often sends you back to framing. Iteration is evidence of care, not failure.

The ordering rules

Frame before touching the data. Understand before cleaning.
Explore before modelling. Evaluate before communicating.

Why the arrows run both ways

Exploration may reveal the target was poorly defined. Evaluation may reveal you need more representative data.

DISCUSS

Why might an evaluation result send you back to the framing or acquisition stage?

What has to be true before a model is useful?

A fast model can still answer the wrong question

Speed and accuracy are properties of the computation, not of the question.

A correct calculation can still be irrelevant

Nothing in the code checks whether anyone needed the number.

Write one sentence, now:

“Before I model, I need to know _____.”

Keep it. We check it against your exit ticket at the end of the session.

★ SHARE

Ask the person beside you: did you name a property of the data, a decision, or an ethical risk?

Storage type is not the same as meaning

Nominal

Categories, no order

programme, study_mode

Ordinal

Ordered, gaps may differ

midterm_satisfaction (1-5)

Discrete

Countable quantities

logins, submissions

Continuous

Any value in a range

entry_score, minutes_on_platform

student_id is nominal

It is stored as an integer, so `describe()` will happily report its mean.

That number is meaningless, and the software will never warn you.

If you can compute a meaningful mean of a column, it is not an identifier.

CHECK

Set identifier columns to str on load. What kind of bug does that one habit prevent?

One row, one meaning

The next twenty minutes are the foundation of Weeks 2 to 5. Nothing here is difficult. It is simply the thing everybody skips.

Rows identify units; columns describe attributes

A dataset is a representation of a process — not the process itself.

One row is one ...

one student
one student-course registration
one student-week of platform activity

The same university, three tables, three different meanings of “a row”.

Columns are variables

programme • study_mode • entry_score
logins • submissions • minutes_on_platform

A column's meaning matters more than the type pandas assigns it on load.

★ SAY IT ALOUD

Before any count, mean, groupby or merge: “One row of this table is one ____.”

One student, three different tables

students.csv

one row = one student

```
student_id | programme | entry_score
100001 | CS | 72
100002 | Psychology | 65
```

1,200 rows

enrolments.csv

one row = one registration

```
student_id | course | grade
100001 | DS100 | 70
100001 | MATH110 | 65
```

4,665 rows

activity.csv

one row = one student-week

```
student_id | week | logins
100001 | 1 | 8
100001 | 2 | 7
```

16,165 rows

Student 100001 appears once in students, twice in enrolments, and once per week in activity.

★ CHOOSE THE TABLE

To count people, which table would you start from — and why?

One row, one meaning

A colleague runs `len(enrolments)` and reports:

Number of students: 4665

The number is wrong, and it is wrong in a way that will never raise an error.

`enrolments` has one row per student-course registration — not one row per student. Each student registers for three or four courses.

Establish what a row is

```
enrolments.student_id.value_counts().head()
```

```
100626  4
100001  4
100394  4
```

```
len(enrolments)          4665
enrolments.student_id.nunique() 1200
```

★ PREDICT

How many distinct students are there? What line of code would you use to check?

Correct code can answer the wrong question

4,665

registration rows

1,200

distinct students

3.9×

overcount

The remedy

If the noun in your question is “student”, aggregate to one row per student before you count.

```
per_student = enrolments.groupby("student_id").agg(n_courses=("course_code", "count"),  
mean_grade=("final_grade", "mean")).reset_index()
```

TURN AND TALK

When would you deliberately want the registration-level table instead of the student-level one?

The wrong unit produces systematic bias, not random noise

One vote per registration

11.9%

```
(merged.study_mode == "Part-time").mean()
```

Wrong — and wrong in a fixed direction.

One vote per student

14.4%

```
(students.study_mode == "Part-time").mean()
```

Right — the unit matches the question.

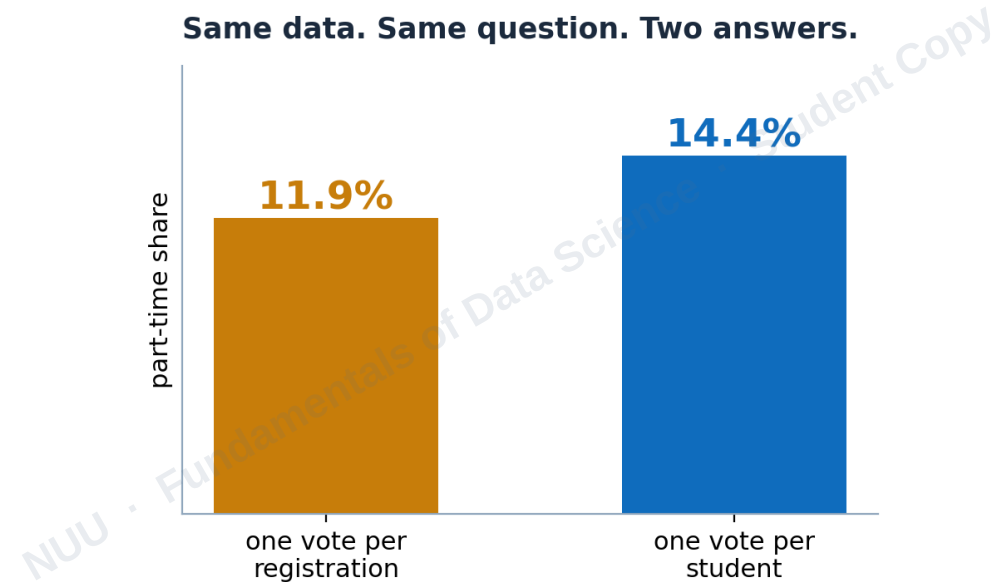
Part-time students defer the heavy quantitative courses, so they hold fewer registrations each — so registration-level counting under-weights them. Anyone checking your arithmetic will find it correct.

★ EXPLAIN

Why is this not a random error that disappears when we collect more data?

Two bars, one of which is wrong

What share of our students study part-time? Both bars come from the same two files, joined the same way. Only the counting unit differs.



An error of 2.5 percentage points, in a fixed direction, that no software will ever flag.

★ BEFORE THE BREAK

If a report reached the Dean with the left-hand number on it, what would be the consequence — and who would notice?

SHORT BREAK — TEN MINUTES

When we return: where data came from changes what it can mean

Before you go, write down one question you still have about:
claims • rows • variables • reproducibility

Come back ready to challenge a number that looks convincing.

SO FAR TODAY

4,665 rows — 1,200 students

11.9% versus 14.4% part-time,
from the same two tables.

*Neither error would raise a single
warning in Python.*

Where the numbers came from

A dataset is not a neutral window onto the world. It is a record made by a system, for its own purposes, with somebody's choices baked in.

Every dataset carries the choices of the system that made it

Files	CSV, JSON, Excel — someone chose the columns
APIs	services expose selected records, on their terms
Databases	operational records kept to run a process
Sensors & surveys	measurement and response choices, both fallible
System exhaust	logs designed for operations — not for your question

Our activity.csv

It exists because the learning platform logs events for operational reasons.

Nobody designed it to measure engagement.

It measures what the platform can see, which is not the same thing.

★ QUESTION

activity.csv records platform activity. Does that mean it measures engagement? What can it not see?

Population, sampling frame, and sample are three different groups

Target population

everyone your conclusion is meant to describe

all enrolled students

Sampling frame

the list you can actually draw from

students with an active account

Sample

the people who end up in your data

those who answered the survey

Coverage error lives in the gaps

Every student who falls between two rings is silently excluded.

No increase in sample size will fix it — a bigger sample just puts a tighter interval around the wrong number.

Reading: Volume 1, §2.3 on sampling strategies. A convenience sample of 5,000 can be less credible than a random 200.

★ APPLY IT

For a platform satisfaction survey, who is excluded before anyone chooses whether to respond?

An operational definition has to be defended, not assumed

Concept	Operational definition	How it can fail
Engagement	submissions in weeks 1–4	measures compliance with deadlines, not learning
Academic difficulty	a stated rule using early evidence	may flag students who need a different kind of support
Satisfaction	survey response on 1–5	non-responders may differ systematically

An operational definition is the rule that turns a concept into a column. Write it in Markdown before you write the code.

★ SMALL GROUP TASK

Give two operational definitions of “engagement”. Which would you trust more, and why?

Reliable is not the same as valid

Reliability

Would the measurement give the same value if taken again?

A login count is highly reliable. It is also only moderately valid — which is the whole point.

Validity

Does it measure the concept, or only something correlated with it?

minutes_on_platform measures a browser tab, not attention.

A proxy stands in for something unmeasured. When a proxy becomes a model input, the model learns the proxy — not the concept — and fails exactly where the two diverge.

CHALLENGE

Name one student behaviour that produces low platform minutes but does not mean disengagement.

Ask who is mismeasured before you operationalise

Compared with full-time students, part-time students show:

–26.5%

minutes on platform

–26.8%

logins

–1.2%

submissions

Same students. Three proxies for “engagement”. Two of them penalise part-time study by roughly a quarter; one barely moves.

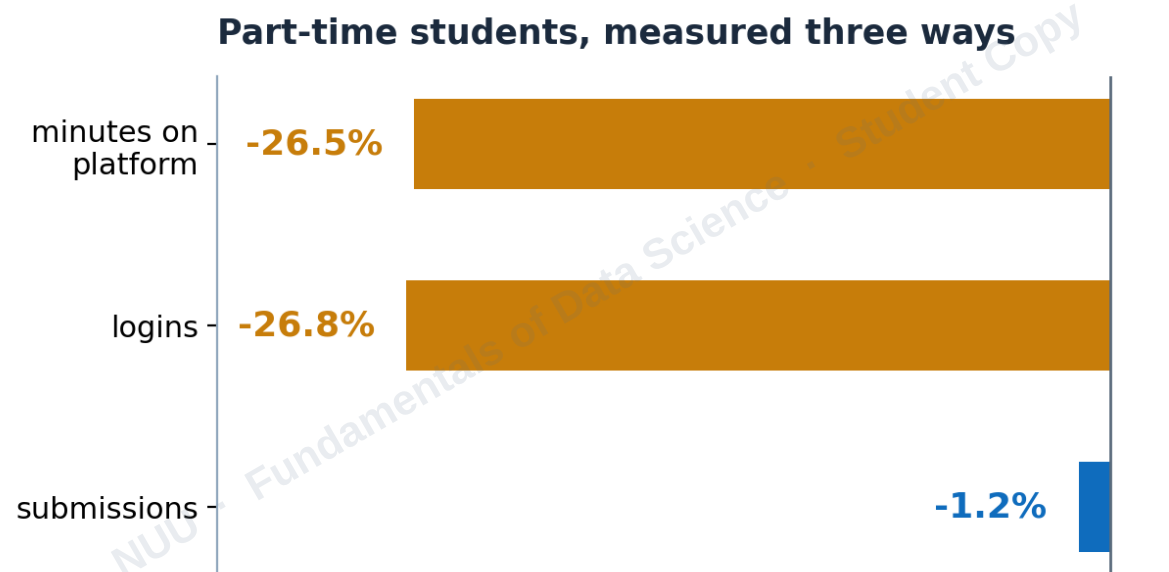
If all three predict withdrawal equally well, submissions are the fairer choice — and the easier one to explain to the student who was flagged.

★ DEBATE

Should a model use the strongest proxy, the fairest proxy, or both? What evidence would settle it?

Three ways to measure the same thing

Part-time students compared with full-time, on each of the three candidate measures of “engagement”.



Two of the three penalise part-time study by about a quarter. One barely moves.

★ CARRY THIS INTO WEEK 10

If the three predict withdrawal equally well, choosing the strongest one over the fairest one is a decision, not a default.

State the expected cardinality before every join

students

one row per student

1,200 rows

enrolments

one row per student-course

4,665 rows

activity

one row per student-week

16,165 rows

Merging enrolments with students: expect many-to-one on student_id. 4,665 rows in, 4,665 rows out. Say the number before you run it.

Joining tables with different units is the single most common way to silently corrupt an analysis.

PREDICT

If a merge turns 4,665 rows into 9,330, what is the most likely data problem?

Let pandas test your merge assumption

```
before = len(enrolments)

merged = enrolments.merge(
    students, on="student_id", how="left",
    validate="many_to_one"
)

assert len(merged) == before
```

validate= raises

if students contains duplicate student IDs — before the join silently doubles your rows.

the assert catches

unintended row growth. Check for unmatched IDs too — a left join hides them as NaN.

validate= is the cheapest bug prevention in pandas. It is required on every merge in this course, from Lab 1 onwards.

READ THE CODE

What assumption does `validate="many_to_one"` make about each table?

A data dictionary and a provenance record are part of the analysis

Data dictionary — per column

name • type • meaning • units
permitted values • missingness • known quirks

The dictionary for this course's data ships with the data. Read it; it is the model for the one you will write.

Provenance — per dataset

source • who collected it • when
consent or licence • what has been done to it since

Neither is bureaucratic overhead.

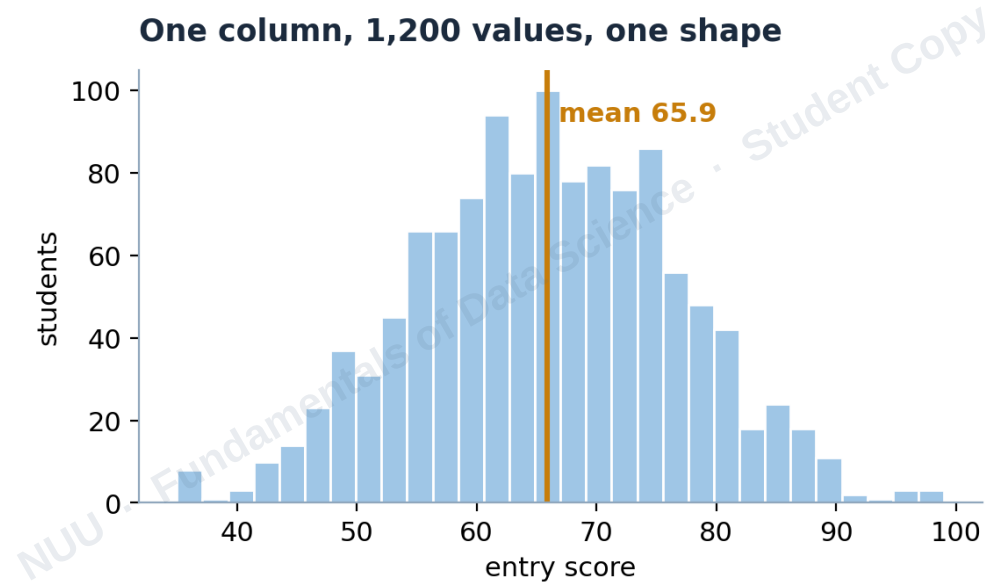
Six weeks from now you will not remember whether passed meant ≥ 50 or ≥ 40 — and whoever inherits your work never knew.

WHY IT MATTERS

Six weeks from now, what will you wish you had written down today?

A column has a shape, and the shape is information

Every entry score in the file, in one picture. You will meet this in Week 2 as a distribution; today it is just proof that a column is more than a list of numbers.



The mean is one number summarising 1,200. The shape tells you whether that number deserves your trust.

WEEK 2 BEGINS HERE

Two datasets can share a mean and share almost nothing else. That is next week's first question.

A notebook is a transparent argument

01	Markdown	states the purpose, the assumptions, and the decisions
02	Code	performs the transformations
03	Output	supplies the evidence
04	Interpretation	connects the evidence back to the question
05	Restart and run all	proves the analysis works from a clean state

The two rules

1. Runtime → Restart and run all before submitting anything.
2. A number without a sentence interpreting it is not an answer.

Stated today. Repeated weekly. Enforced in marking from Week 4.

★ SELF-CHECK

Could a classmate restart your notebook, understand every output, and reach the same conclusion?

Ethics shapes the question before the model exists

Before collecting or using data about people, ask:

Who benefits?

Who bears the risk if we are wrong?

Is participation voluntary?

Could a less intrusive analysis meet the purpose?

Support, audit, or decide?

A variable can be right for one and wrong for another.

Can it be contested?

Can the affected person understand and challenge the outcome?

first_generation may be useful for auditing unequal outcomes and still be inappropriate as a decision feature. That distinction runs through the whole course, and it is graded.

★ COMMITMENT

Finish the sentence: "A prediction of difficulty should be used to _____, not to _____."

Before you go

Three things I need from you, and two things you need from me.

Ungraded — this tells me how to teach Weeks 2 to 4

Write a function that reads a CSV file and returns the mean of one named column.

On paper. No marks. No names if you prefer.

It is fine if you cannot finish it. It is fine if you have never written a function. What I need is an honest picture of the room, and I would much rather have it in Week 1 than in Week 4.

What this is for

“Somewhat familiar with programming” can mean loops and functions, or it can mean having seen one for-loop.

Those two rooms need different amounts of scaffolding in Weeks 2 to 4.

I will adjust the labs based on what comes back — which only works if you answer honestly.

★ FIVE MINUTES

Hand it in on the way out, with the Lab 0 sheet.

What you can do now that you could not at 9am

A concern is not a question

until it names a population, a variable, a period and a decision

Evidence has a ceiling

a correlation supports a description, never an intervention

A third variable can explain both

entry score predicted logins and grades before either happened

A row is not a thing

it is a choice, and 4,665 registrations are not 4,665 students

The wrong unit biases in a direction

11.9% against 14.4%, and no warning either way

A proxy decides who is mismeasured

minutes and logins penalise part-time study; submissions do not

Not one of these is about an algorithm. All six are about what a number is allowed to mean.

[BACK TO THE GALLERY](#)

Row 03 of the pattern — “someone decided what a row meant” — was the whole of today.

Take these into the lab

unit of observation

what one row of a table represents

confounder

a third variable that explains an association

sampling frame

the list you can actually draw from

operational definition

the rule that turns a concept into a column

reliability

would it give the same value if measured again

cardinality

how many rows on each side of a join

claim ladder

descriptive → comparative → predictive → causal

target population

everyone your conclusion is meant to describe

coverage error

who falls between the population and the frame

proxy

a measurable stand-in for something you cannot measure

validity

does it measure the concept, or only something nearby

provenance

where the data came from and what was done to it

ALL TWELVE ARE IN THE GLOSSARY

Volume 2, at the back. You are not expected to memorise them — you are expected to use them.

Show your Week 1 thinking

Four short answers before you leave:

- 1 Turn “students are struggling” into an analysable question.
- 2 Label one claim: descriptive, comparative, predictive, or causal.
- 3 Finish the sentence: “One row of enrolments.csv is one _____.”
- 4 Name one limitation or fairness risk in using platform logs as evidence.

Bring these answers to Lab 1 — then test your claims directly, in a reproducible notebook.

★ **HAND IN**

Exit ticket, diagnostic, and your name on the Lab 0 sheet if the setup cell did not run.

What to read, and what is coming

Read — about ninety minutes

Volume 1, Chapters 1 and 2 in full.

Including §2.3 on sampling strategies, which we set as reading today rather than covering in class.

Do — about two hours

Lab 1. Bring your exit-ticket answers with you; the lab tests them against the actual files.

Week 2 asks a question today could not:

“The mean entry score is 65.9.” Fine — but is that one number a fair summary of 1,200 people, or is it hiding two different groups?

★ ONE MORE THING

If your Lab 0 setup cell did not run today, your name is on the sheet. I will find you this week.

Your first transparent inspection

Before the lab

Seven CSV files in MyDrive/fds/data/, and the setup cell returning the four shapes.

Deliverable

One notebook plus a 150-word reflection naming one limitation you would flag to a decision-maker.

In the lab you will

- Load the four tables and state in Markdown what one row means in each — then verify it in code
- Check that `student_id` is unique, and that every student has the number of weeks you expect
- Write one descriptive question, one predictive question, and one question this dataset cannot answer
- Restart and run all, confirm every output is identical, then submit

Carry this rule with you: before any count, mean, groupby or merge, state the unit of observation.